Typos List:

- On page 24 above equation 6, below "the above gives us" shouldn't the randomness be over ξ as well as z
- Page 42 claims that the definition of $\mathcal{E}_{\text{test}}$ is that $\left|\boldsymbol{w}_{\perp}^{\top}\boldsymbol{x}\right| < C\|\boldsymbol{w}_{\perp}\|\log^{1/2}(d)$ but really this is just part of the definition/implied, $\mathcal{E}_{\text{test}}$ also includes a bound on flipping the sign for any x_i where $i > 3$
- Page 87 lists F.15 as a Theorem but in the appendix it's a lemma

Questions List:

- How would you go about discovering the assumptions used in the inductive hypothesis for each recurrence? It's clear how recurrences imply asymptotics and how the scalings follow from constraints from doing the proofs, but it's not obvious to me how we find the inductive hypothesis in the first place such that we can show it's maintained

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Setup Definitions

Data

- Data $\boldsymbol{x}$ is distributed like i.i.d Rademacher in all entries except the third entry, and the target function is $y(x) = -x_1 x_2$, the XOR on the first two entries
- The third entry is like a planted answer, with probability $1 - \lambda$ it is the XOR of the first two entries, and with probability λ it's the negation
- We decompose $\boldsymbol{w} = \boldsymbol{w}_{\text{sig}} + \boldsymbol{w}_{\text{opp}} + \boldsymbol{w}_{\text{sp}} + \boldsymbol{w}_{\perp}$ where sig is the projection onto the signal direction, which depends on the sign of a, $\boldsymbol{w}_{\text{opp}}$ points in the "other" direction (discussed below), $\boldsymbol{w}_{\text{sp}}$ is the projection onto the third coordinate, and $\boldsymbol{w}_{\text{perp}}$ is everything else
  - The sign of a determines the sign of a particular neuron's contribution, and therefore whether we want to align with $\pm\mu_1$ (the choices of $\boldsymbol{z}$ that are 1 under XOR) vs $\pm\mu_2$ (the -1 inputs) so $\boldsymbol{w}_{\text{opp}}$ is just the orthogonal direction to the signal
- We decompose $\boldsymbol{x} = \boldsymbol{z} + \boldsymbol{s} + \boldsymbol{\xi}$, which are the projections onto the first two coordinates (so $y(\boldsymbol{x}) = y(\boldsymbol{z})$), the projection onto the third coordinate, and the remaining $d - 3$ coordinates respectively

Loss

- The neural network output $f_\rho(x)$ is the average of the neuron outputs, so

$$f_\rho(\boldsymbol{x}) = \frac{1}{p} \sum_{a_j, w_j} a_j \sigma(\boldsymbol{w}^\top \boldsymbol{x}) = E_{(a_j, \boldsymbol{w}_j) \sim \rho} \left[a_j \sigma(\boldsymbol{w}^\top \boldsymbol{x}) \right]$$

- The loss, as a function of $\boldsymbol{x}$ denoted $\ell_\rho(\boldsymbol{x})$, is twice the negative log of the sigmoid of the margin, where we let $\gamma(\boldsymbol{x}) \coloneqq f_\rho(x) y(x)$ denote the margin, $\psi(\gamma) = \frac{1}{1+\exp(-\gamma)}$ denote the sigmoid, and so $\ell_\rho(x) = -2\log(\psi(\gamma(x))) = h(\gamma(x))$ where we use h to denote the outer functions, so $h(\gamma) \coloneqq -2\log(\psi(\gamma))$ is the sigmoid composed with twice the negative log.
- We let $\ell_0(x)$ denote the first order approximation of ℓ_ρ around 0 with respect to $f_\rho(x)$, observe that because h is scaled by 2 we have $h'(\gamma) = \frac{-2e^{-\gamma}}{1+e^{-\gamma}}$ so $h'(0) = -1$, giving

$$\ell_0\big(f_\rho(\boldsymbol{x})\big) = \ell_\rho(0) + h'(0)\gamma(\boldsymbol{x}) = -2\log\frac{1}{2} - y(\boldsymbol{x}) f_\rho(\boldsymbol{x})$$

- L_ρ and L_0 are the corresponding population losses, so the expectation over $\boldsymbol{x}$ drawn from the described distribution
- We write $\ell_\rho^{(1)}(\boldsymbol{x}) = h'(\gamma(\boldsymbol{x}))$ for the first derivative of the loss with respect to the margin, and $\ell_0^{(1)}(\boldsymbol{x})$ as the corresponding value for ℓ_0 which is -1. So I believe we're writing it as a function of $\boldsymbol{x}$ but really are considering the change in the loss wrt the margin, later this will be useful in bounding $\nabla\big(L_\rho - L_0\big)$.
- $\nabla_{\boldsymbol{w}} L$ and $\partial_a L$ are scaled by p so that we get invariance over the width of the neural network (the p cancels with the $\frac{1}{p}$ in the averaging over neurons)

Section B: L_0 Analysis

We want to characterize the gradients under the L_0 approximation of the loss to derive recurrences on each part of $\boldsymbol{w}$. We will ultimately show that the L_0 approximation is fine for small networks (small absolute value/norm of $(a, \boldsymbol{w})$), and that both $\boldsymbol{w}_{\text{sig}}, \boldsymbol{w}_{\text{sp}}$ grow according to a geometric recurrence, but that $\boldsymbol{w}_{\text{sp}}$ has a headstart like $\sqrt{d}$.

Lemma 0.1 (L_0 gradient along signal and opposite directions): For any neuron $(a, \boldsymbol{w})$ we have

$$-\boldsymbol{w}_{\text{sig}}^\top \nabla_{\boldsymbol{w}} L_0 = \frac{\sqrt{2}}{4}|a|\|\boldsymbol{w}_{\text{sig}}\| \Pr_{\boldsymbol{\xi}}\left[|\boldsymbol{w}^\top \boldsymbol{e}_3 + \boldsymbol{w}^\top \boldsymbol{\xi}| \le \sqrt{2}\|\boldsymbol{w}_{\text{sig}}\|\right]$$

$$-\boldsymbol{w}_{\text{opp}}^\top \nabla_{\boldsymbol{w}} L_0 = \frac{\sqrt{2}}{4}|a|\|\boldsymbol{w}_{\text{opp}}\| \Pr_{\boldsymbol{\xi}}\left[|\boldsymbol{w}^\top \boldsymbol{e}_3 + \boldsymbol{w}^\top \boldsymbol{\xi}| \le \sqrt{2}\|\boldsymbol{w}_{\text{opp}}\|\right]$$

Proof of Lemma 0.1: Note that we have by definition of L_0 as the first order approximation and a previous calc that

$$\begin{aligned}
-\boldsymbol{w}_{\text{sig}}^\top \nabla_w L_0 &= -\boldsymbol{w}_{\text{sig}}^\top \mathbb{E}\left[p\frac{\partial}{\partial \boldsymbol{w}} - y(\boldsymbol{x}) f_\rho(\boldsymbol{x})\right] \\
&= \boldsymbol{w}_{\text{sig}}^\top \mathbb{E}\left[y(x) p \frac{\partial}{\partial \boldsymbol{w}} \frac{1}{p}\sum_j a_j \sigma(\boldsymbol{w}^\top \boldsymbol{x})\right] \\
&= \boldsymbol{w}_{\text{sig}}^\top \mathbb{E}[y(x) a \sigma'(\boldsymbol{w}^\top \boldsymbol{x})\boldsymbol{x}]
\end{aligned}$$

and by recognizing that $\boldsymbol{w}_{\text{sig}}^\top$ is orthogonal to $\boldsymbol{x} - \boldsymbol{z}$ we can split the above by using $\boldsymbol{x} = \boldsymbol{z} + \boldsymbol{s} + \boldsymbol{\xi}$ so that the above equals $\mathbb{E}\left[y(\boldsymbol{z}) a \sigma'(\boldsymbol{w}^\top \boldsymbol{x}) \boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}\right]$ also using that y is determined by $\boldsymbol{z}$. So intuitively this part gradient only depends on $\boldsymbol{x}$ in units $3, ..., d$ via this inner sigmoid, as everything else is projected into μ_{sig}'s span. From here we use iterated expectations to condition on whether $\boldsymbol{w}_{\text{sig}} \parallel \boldsymbol{z}$, noting that the above is zero when the two are orthogonal to get

$$\begin{aligned}
&= \frac{1}{2} \mathop{\mathbb{E}}_{\substack{x,z \\ \boldsymbol{w}_{\text{sig}} \parallel \boldsymbol{z}}} \left[y(z) a \sigma'(\boldsymbol{w}^\top \boldsymbol{x}) \boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}\right] \\
&= \frac{1}{2} \mathop{\mathbb{E}}_{\substack{x,z \\ \boldsymbol{w}_{\text{sig}} \parallel \boldsymbol{z}}} \left[y(z) a \sigma'\left(\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{s} + \boldsymbol{w}^\top \boldsymbol{\xi}\right) \boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}\right]
\end{aligned}$$

letting $\boldsymbol{z}_0$ denote one of the two values of $\boldsymbol{z}$ such that we have $\boldsymbol{w}_{\text{sig}} \parallel z_0$ (so if $a > 0$ then μ_1 else μ_2) we know that $(\boldsymbol{z}, \boldsymbol{s})$ are independent and $\boldsymbol{z}$ is $\pm \boldsymbol{z}_0$ with probability $1/2$ each, and that $\boldsymbol{s}$ is $\boldsymbol{s}_{\text{maj}}$ with prob $1 - \lambda$ and $\boldsymbol{s}_{\text{min}}$ with prob λ. Recall that $\boldsymbol{s}_{\text{maj}}$ is just aligned with the sign of a and so we have $\boldsymbol{s} = \boldsymbol{s}_{\text{maj}}$ with prob $1 - \lambda$ because $s = \boldsymbol{s}_{\text{maj}}$ if and only if s and a share signs, which is the same as s being one for $z = \pm \mu_1$ and -1 otherwise. **TODO:** Okay I should review this at some point since walking back how things are defined relative to sign of a seems difficult to me, there's probably an easier way to see it, any way this allows us to expand the above. This is probably the same as just thinking of $\boldsymbol{s}_{\text{maj}}$ as the xor of $\boldsymbol{z}_0$ but that's not the definition itself.

Noting that we're just varying the signs of $\boldsymbol{z}, \boldsymbol{s}$ let's introduce the shorthand $\sigma'(\pm, \pm)$ to mean $\sigma'\left(\boldsymbol{w}_{\text{sig}}^\top(\pm \boldsymbol{z}) + \boldsymbol{w}^\top(\pm \boldsymbol{s}) + \boldsymbol{w}^\top \boldsymbol{\xi}\right)$ so the above equals

$$
\begin{aligned}
&= \frac{1}{2}a\Big(\\
&\left(\frac{1}{2} - \frac{\lambda}{2}\right) \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[y(\boldsymbol{z}_0)\sigma'(+,+)\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\right] \\
&\quad + \frac{\lambda}{2} \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[y(\boldsymbol{z}_0)\sigma'(+,-)\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\right] \\
&+ \left(\frac{1}{2} - \frac{\lambda}{2}\right) \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[y(-\boldsymbol{z}_0)\sigma'(-,+)\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}_0)\right] \\
&\quad + \frac{\lambda}{2} \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[y(-\boldsymbol{z}_0)\sigma'(-,-)\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}_0)\right] \\
&\quad \Big)
\end{aligned}
$$

and we can factor our the $1/2$, then observe that $y(\boldsymbol{z}_0) = y(-\boldsymbol{z}_0)$ with $ay(\boldsymbol{z}_0) = ay(-\boldsymbol{z}_0) = |a|$ by the fact $\boldsymbol{z}_0$'s xor aligns with the sign of a to get

$$
\begin{aligned}
&= \frac{1}{4}|a|\big(\\
&(1-\lambda) \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[\sigma'(+,+)\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\right] \\
&\quad + \lambda \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[\sigma'(+,-)\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\right] \\
&+ (1-\lambda) \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[\sigma'(-,+)\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}_0)\right] \\
&\quad + \lambda \cdot \mathbb{E}_{\boldsymbol{\xi}}\left[\sigma'(-,-)\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}_0)\right]\big) \\
= \frac{1}{4}|a|&\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\big(\\
&(1-\lambda) \cdot \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(+,+)] \\
&\quad + \lambda \cdot \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(+,-)] \\
&- (1-\lambda) \cdot \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(-,+)] \\
&\quad - \lambda \cdot \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(-,-)]\big)
\end{aligned}
$$

from here we'll group terms according to the sign of $\boldsymbol{s}_{\text{maj}}$ (this will let us apply an identity about differences of certain indicators) to get

$$
\begin{aligned}
= \frac{1}{4}|a|&\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\big(\\
&\quad \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(+,+) - \sigma'(-,+)] \\
&+ \lambda \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(-,+) - \sigma'(+,+)] \\
&+ \lambda \mathbb{E}_{\boldsymbol{\xi}}[\sigma'(+,-) - \sigma'(-,-)]\big)
\end{aligned}
$$

unrolling definitions now since the manipulations are done and recognizing $\sigma'(x) = \mathbb{1}[x \geq 0]$, the above is

$$= \frac{1}{4}|a|\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0 \Big(\\ \mathbb{E}_{\boldsymbol{\xi}}\Big[\mathbb{1}\Big[\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{s} + \boldsymbol{w}^\top \boldsymbol{\xi}\Big] - \mathbb{1}\Big[\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}) + \boldsymbol{w}^\top \boldsymbol{s} + \boldsymbol{w}^\top \boldsymbol{\xi}\Big]\Big] \\ +\lambda \mathbb{E}_{\boldsymbol{\xi}}\Big[\mathbb{1}\Big[\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}) + \boldsymbol{w}^\top \boldsymbol{s} + \boldsymbol{w}^\top \boldsymbol{\xi}\Big] - \mathbb{1}\Big[\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{s} + \boldsymbol{w}^\top \boldsymbol{\xi}\Big]\Big] \\ +\lambda \mathbb{E}_{\boldsymbol{\xi}}\Big[\mathbb{1}\Big[\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z} + \boldsymbol{w}^\top (-\boldsymbol{s}) + \boldsymbol{w}^\top \boldsymbol{\xi}\Big] - \mathbb{1}\Big[\boldsymbol{w}_{\text{sig}}^\top (-\boldsymbol{z}) + \boldsymbol{w}^\top (-\boldsymbol{s}) + \boldsymbol{w}^\top \boldsymbol{\xi}\Big]\Big]\Big)$$

From here consider that for $a, b, c \in \mathbb{R}$ we have

$$\mathbb{1}[a + b + c \geq 0] - \mathbb{1}[-1 + b + c \geq 0] = \mathbb{1}[b + c \geq -a] - \mathbb{1}[b + c \geq a]$$

where the above is 1 if and only if the first indicator is true and the second false, and -1 if the first false and the second true, and 0 otherwise. Therefore the above is

$$= \begin{cases} 1 & \text{when } a \leq b + c \leq -a \\ -1 & \text{when } -a \leq b + c \leq a \\ 0 & \text{else} \end{cases}$$

or equivalently, this is nonzero when $|b + c| \leq |a|$ with the sign being the opposite of a's sign as the sign of a determines whether which case is valid. This type of argument works for all three differences of indicators and actually lets us group the first two differences of indicators since they share $\boldsymbol{w}^\top \boldsymbol{s} + \boldsymbol{w}^\top \boldsymbol{\xi}$ just up to a sign flip so the above equals

$$= \frac{1}{4}|a|\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0 \Big(\\ (1-\lambda) \Pr_{\boldsymbol{\xi}}\Big[\big|\boldsymbol{w}^\top \boldsymbol{s}_{\text{maj}} + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \big|\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\big|\Big] \operatorname{sgn}\Big(\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\Big) \\ +\lambda \Pr_{\boldsymbol{\xi}}\Big[\big|-\boldsymbol{w}^\top \boldsymbol{s}_{\text{maj}} + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \big|\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\big|\Big] \operatorname{sgn}\Big(\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\Big)\Big).$$

We now observe that $\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0 \operatorname{sgn}\Big(\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\Big) = \big|\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\big| = \big\|\boldsymbol{w}_{\text{sig}}^\top\big\|\|z_0\| = \sqrt{2}\big\|\boldsymbol{w}_{\text{sig}}^\top\big\|$ with the last two steps following from the fact $\boldsymbol{w}_{\text{sig}} \parallel \boldsymbol{z}_0$ and so Cauchy-Schwarz is tight here, with $\|\boldsymbol{z}_0\| = \sqrt{2}$ following from definition of $\boldsymbol{z}_0$. We use this to factor out $\operatorname{sgn}\Big(\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\Big)$ from both terms and also to rewrite the $\big|\boldsymbol{w}_{\text{sig}}^\top \boldsymbol{z}_0\big|$ term inside each probability:

$$= \frac{\sqrt{2}}{4}|a|\big\|\boldsymbol{w}_{\text{sig}}\big\| \Big(\\ (1-\lambda) \Pr_{\boldsymbol{\xi}}\Big[\big|\boldsymbol{w}^\top \boldsymbol{s}_{\text{maj}} + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \sqrt{2}\big\|\boldsymbol{w}_{\text{sig}}\big\|\Big] \\ +\lambda \Pr_{\boldsymbol{\xi}}\Big[\big|-\boldsymbol{w}^\top \boldsymbol{s}_{\text{maj}} + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \sqrt{2}\big\|\boldsymbol{w}_{\text{sig}}\big\|\Big]\Big).$$

and we conclude the proof by noting that $\boldsymbol{\xi}$ is symmetric, so in particular we can flip $\boldsymbol{\xi}$ with $-\boldsymbol{\xi}$ while preserving probabilities; because of the absolute value this is the same as flipping the sign of $\boldsymbol{s}_{\text{maj}}$ so

$$\Pr_{\boldsymbol{\xi}}\Big[\big|-\boldsymbol{w}^\top \boldsymbol{s}_{\text{maj}} + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \sqrt{2}\big\|\boldsymbol{w}_{\text{sig}}\big\|\Big] = \Pr_{\boldsymbol{\xi}}\Big[\big|\boldsymbol{w}^\top \boldsymbol{s}_{\text{maj}} + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \sqrt{2}\big\|\boldsymbol{w}_{\text{sig}}\big\|\Big] = \\ \Pr_{\boldsymbol{\xi}}\Big[\big|\boldsymbol{w}^\top \boldsymbol{e}_3 + \boldsymbol{w}^\top \boldsymbol{\xi}\big| \leq \sqrt{2}\big\|\boldsymbol{w}_{\text{sig}}\big\|\Big]$$

which allows us to conclude the above equals

$$= \frac{\sqrt{2}}{4}|a|\|\boldsymbol{w}_{\text{sig}}\| \Pr_{\boldsymbol{\xi}}\left[|\boldsymbol{w}^\top \boldsymbol{e}_3 + \boldsymbol{w}^\top \boldsymbol{\xi}| \le \sqrt{2}\|\boldsymbol{w}_{\text{sig}}\|\right]$$

as desired. ■

Lemma 0.2 (L_0 gradient along spurious direction): For any neuron $(a, \boldsymbol{w})$ we have

$$-\boldsymbol{w}_{\text{sp}}^\top \nabla_{\boldsymbol{w}} L_0 = a w_{\text{sp}}\left(\frac{1}{2} + \frac{\varepsilon_{\boldsymbol{w}}}{4} - \lambda\right)$$

where

$$A = \min(\{\pm \boldsymbol{w}^\top \boldsymbol{\mu}_1, \pm \boldsymbol{w}^\top \boldsymbol{\mu}_2\}), \quad B = \min(\{\pm \boldsymbol{w}^\top \boldsymbol{\mu}_1, \pm \boldsymbol{w}^\top \boldsymbol{\mu}_2\} \setminus A)$$

$$\varepsilon_{\boldsymbol{w}} = \begin{cases} \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} - w_{\text{sp}} \in [A, B]\right] - \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \in [A, B]\right] & \text{if} \operatorname{sgn}(w_1) = \operatorname{sgn}(w_2) \\ \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \in [A, B]\right] - \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} - w_{\text{sp}} \in [A, B]\right] & \text{if} \operatorname{sgn}(w_1) \neq \operatorname{sgn}(w_2) \end{cases}$$

Proof of Lemma 0.2: Begin as before using that $\nabla_{\boldsymbol{w}} L_0 = \mathbb{E}_x[-y(z) a \sigma'(\boldsymbol{w}^\top \boldsymbol{x})\boldsymbol{x}]$ to get

$$-\boldsymbol{w}_{\text{sp}}^\top \nabla_{\boldsymbol{w}} L_0 = \mathbb{E}_{\boldsymbol{x}}\left[y(\boldsymbol{z}) a \sigma'(\boldsymbol{w}^\top \boldsymbol{x}) \boldsymbol{w}_{\text{sp}}^\top \boldsymbol{s}\right]$$

and then expand over $\boldsymbol{s}$ using that $\boldsymbol{s} = y(\boldsymbol{z})$ with probability $1 - \lambda$, with $\boldsymbol{x}, \boldsymbol{\xi}$ independent of $\boldsymbol{s}$ to get

$$\begin{aligned} = a\big((1-\lambda)\mathbb{E}_{\boldsymbol{z},\boldsymbol{\xi}}\left[y(\boldsymbol{z})\sigma'(\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top y(\boldsymbol{z}) \boldsymbol{e}_3 + \boldsymbol{w}^\top \boldsymbol{\xi}) \boldsymbol{w}_{\text{sp}}^\top y(\boldsymbol{z}) \boldsymbol{e}_3\right] \\ + \lambda \mathbb{E}_{\boldsymbol{z},\boldsymbol{\xi}}\left[y(\boldsymbol{z})\sigma'(\boldsymbol{w}^\top \boldsymbol{z} - \boldsymbol{w}^\top y(\boldsymbol{z}) \boldsymbol{e}_3 + \boldsymbol{w}^\top \boldsymbol{\xi}) \boldsymbol{w}_{\text{sp}}^\top (-y(\boldsymbol{z}) \boldsymbol{e}_3)\right]\big) \end{aligned}$$

and we factor out $y(\boldsymbol{z})^2 = 1$, rewriting expectations of indicators as probabilities (using again that the derivative of the relu is an indicator) to get

$$\begin{aligned} = a w_{\text{sp}}\Big((1-\lambda) \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge -y(\boldsymbol{z}) w_{\text{sp}}\right] \\ + \lambda \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge y(\boldsymbol{z}) w_{\text{sp}}\right]\Big) \end{aligned}$$

from here observe that by symmetry of $\boldsymbol{z}$ and the fact that $y(\boldsymbol{z}) = y(-\boldsymbol{z})$ followed by using the symmetry of $\boldsymbol{\xi}$ we have

$$\begin{aligned} &\Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge -y(\boldsymbol{z}) w_{\text{sp}}\right] + \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge y(\boldsymbol{z}) w_{\text{sp}}\right] \\ &= \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge -y(\boldsymbol{z}) w_{\text{sp}}\right] + \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[-\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge y(\boldsymbol{z}) w_{\text{sp}}\right] \\ &= \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge -y(\boldsymbol{z}) w_{\text{sp}}\right] + \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[-\boldsymbol{w}^\top \boldsymbol{z} - \boldsymbol{w}^\top \boldsymbol{\xi} \ge y(\boldsymbol{z}) w_{\text{sp}}\right] \\ &= \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge -y(\boldsymbol{z}) w_{\text{sp}}\right] + \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} < -y(\boldsymbol{z}) w_{\text{sp}}\right] = 1 \end{aligned}$$

I'm not being careful here with strict inequalities vs including the equals case but I don't think it matters. This gives that our desired quantity equals

$$= a w_{\text{sp}}\left(\Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \ge -y(\boldsymbol{z}) w_{\text{sp}}\right] - \lambda\right)$$

so it remains to show that the above probability equals $\frac{1}{2} + \frac{\varepsilon_w}{4}$. We first expand over the distribution of $z \in \{\pm\mu_1, \pm\mu_2\}$ unifomrly to get

$$\begin{aligned}\Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{z} + \boldsymbol{w}^\top \boldsymbol{\xi} \geq -y(\boldsymbol{z})\boldsymbol{w}_{\text{sp}}\right] &= \Pr_{\boldsymbol{z},\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + y(\boldsymbol{z})w_{\text{sp}} > -\boldsymbol{w}^\top \boldsymbol{\mu}_1\right] \\ &= \frac{1}{4}\Big(\Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} > -\boldsymbol{w}^\top \boldsymbol{\mu}_1\right] \\ &\quad + \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} > \boldsymbol{w}^\top \boldsymbol{\mu}_1\right] \\ &\quad + \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} - w_{\text{sp}} > -\boldsymbol{w}^\top \boldsymbol{\mu}_2\right] \\ &\quad + \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} - w_{\text{sp}} > \boldsymbol{w}^\top \boldsymbol{\mu}_2\right]\Big)\end{aligned}$$

and then take complements for the first two probabilities, and use the symmetry of $\boldsymbol{\xi}$ for the last two to rewrite the above as

$$\begin{aligned}= \frac{1}{2} + \frac{1}{4}\Big(&-\Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \leq -\boldsymbol{w}^\top \boldsymbol{\mu}_1\right] \\ &- \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \leq \boldsymbol{w}^\top \boldsymbol{\mu}_1\right] \\ &+ \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \leq \boldsymbol{w}^\top \boldsymbol{\mu}_2\right] \\ &+ \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \leq -\boldsymbol{w}^\top \boldsymbol{\mu}_2\right]\Big)\end{aligned}$$

let $A \leq B \leq C \leq D$ be the ordering of $\{\pm\boldsymbol{w}^\top \boldsymbol{\mu}_1, \pm\boldsymbol{w}^\top \boldsymbol{\mu}_2\}$, observing that $A = -D$ and $B = -C$ with $A, B \leq 0$ and $C, D \geq 0$. Therefore the ordering is determined by whether $|A|, |B|$ is bigger, which is equivalent to whether $\text{sgn}(w_1) = \text{sgn}(w_2)$ as $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ subtract and add these two entries respectively when taking inner products.

Considering first the case where $\text{sgn}(w_1) = \text{sgn}(w_2)$ and so $\boldsymbol{w}^\top \boldsymbol{\mu}_2$ is larger in absolute value write $X = \boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}}$; we have that the above probability is

$$\begin{aligned}&-\Pr[X \leq B] - \Pr[X \leq C] + \Pr[X \leq A] + \Pr[X \leq D] \\ &= \Pr[X \leq A] + \Pr[X \leq D] - \Pr[X \leq B] - \Pr[X \leq C] \\ &= \Pr[X \leq A] + (\Pr[X \leq A] + \Pr[A \leq X \leq D]) \\ &-(\Pr[X \leq A] + \Pr[A \leq X \leq B]) - (\Pr[X \leq A] + \Pr[A \leq X \leq C)] \\ &= \Pr[A \leq X \leq D] - \Pr[A \leq X \leq B] - \Pr[A \leq X \leq C] \\ &= \Pr[C \leq X \leq D] - \Pr[A \leq X \leq B]\end{aligned}$$

and we notice that the above is $\Pr\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \in [C, D]\right] - \Pr\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \in [A, B]\right]$ where the symmetry of $\boldsymbol{\xi}$ gives that the first term equals $\Pr\left[-\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \in [C, D]\right] = \Pr\left[\boldsymbol{w}^\top \boldsymbol{\xi} - w_{\text{sp}} \in [A, B]\right]$ where we used that $A = -C, B = -D$. Therefore as desired, the above equals

$$\Pr\left[\boldsymbol{w}^\top \boldsymbol{\xi} - w_{\text{sp}} \in [A, B]\right] - \Pr\left[\boldsymbol{w}^\top \boldsymbol{\xi} + w_{\text{sp}} \in [A, B]\right]$$

In the case where $\text{sgn}(w_1) \neq \text{sgn}(w_2)$ you can check that we get the negation of the original expression with respect to the ordering of A, B, C, D so the same argument applies with the result being the negation of the above probability, as claimed. ■

Lemma 0.3 ($\varepsilon_{\boldsymbol{w}}$ is exponentially small): For $\varepsilon_{\boldsymbol{w}}$ defined as in Lemma 0.2 we have

$$|\varepsilon_{\boldsymbol{w}}| \lesssim \frac{\|\boldsymbol{w}_{1:2}\|}{\|\boldsymbol{w}_\perp\|} \exp\left(\frac{\sqrt{2}\|\boldsymbol{w}_{1:2}\|\|\boldsymbol{w}_{\text{sp}}\| - w_{\text{sp}}^2}{\|\boldsymbol{w}_\perp\|_2^2}\right) + \frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

Proof of Lemma 0.3: Note that for bounding the absolute value, by the definition of $\varepsilon_{\boldsymbol{w}}$, we can take the first case without loss of generality such that $\varepsilon_{\boldsymbol{w}} = \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top\boldsymbol{\xi} - w_{\text{sp}} \in [A, B]\right] - \Pr_{\boldsymbol{\xi}}\left[\boldsymbol{w}^\top\boldsymbol{\xi} + w_{\text{sp}} \in [A, B]\right]$ as the other case has the same absolute value.

From here the high level appraoch is use a form of Berry-Esseen given in **TODO:** appendix to rewrite the above as a difference of Gaussian cdfs and apply a few identities to get our desired bound. Writing $\varphi(u)$ for the pdf of a Gaussian with mean zero and variance $\|\boldsymbol{w}_\perp\|^2$, Berry-Esseen gives that

$$\begin{aligned}
\varepsilon_{\boldsymbol{w}} &\lesssim \int_{A+w_{\text{sp}}}^{B+w_{\text{sp}}} \varphi(u)du - \int_{A-w_{\text{sp}}}^{B-w_{\text{sp}}} \varphi(u)du + \frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|^{2^3}} \\
&= \int_A^B \big(\varphi(u + w_{\text{sp}}) - \varphi(u - w_{\text{sp}})\big)du + \frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}
\end{aligned}$$

where the definition of the Gaussian pdf gives that the integral term is

$$\begin{aligned}
&\int_A^B \frac{1}{\|\boldsymbol{w}_\perp\|\sqrt{2\pi}}\Big(e^{-(u+w_{\text{sp}})^2/2\|\boldsymbol{w}_\perp\|^2} - e^{-(u-w_{\text{sp}})^2/2\|\boldsymbol{w}_\perp\|^2}\Big)du \\
&= \int_A^B \frac{1}{\|\boldsymbol{w}_\perp\|\sqrt{2\pi}}\Big(e^{(-u^2-2uw_{\text{sp}}-w_{\text{sp}}^2)/2\|\boldsymbol{w}_\perp\|^2} - e^{(-u^2+2uw_{\text{sp}}-w_{\text{sp}}^2)/2\|\boldsymbol{w}_\perp\|^2}\Big)du \\
&= \int_A^B \frac{1}{\|\boldsymbol{w}_\perp\|\sqrt{2\pi}} e^{(-u^2-w_{\text{sp}}^2)/2\|\boldsymbol{w}_\perp\|^2}\Big(e^{-2uw_{\text{sp}}/2\|\boldsymbol{w}_\perp\|^2} - e^{2uw_{\text{sp}}/2\|\boldsymbol{w}_\perp\|^2}\Big)du \\
&= \int_A^B \left(\frac{1}{\|\boldsymbol{w}_\perp\|\sqrt{2\pi}} e^{-w_{\text{sp}}^2/2\|\boldsymbol{w}_\perp\|^2}\right) e^{-u^2/2\|\boldsymbol{w}_\perp\|^2}\Big(e^{-2uw_{\text{sp}}/2\|\boldsymbol{w}_\perp\|^2} - e^{2uw_{\text{sp}}/2\|\boldsymbol{w}_\perp\|^2}\Big)du \\
&= \varphi(w_{\text{sp}}) \int_A^B e^{-u^2/2\|\boldsymbol{w}_\perp\|^2}\Big(e^{-uw_{\text{sp}}/\|\boldsymbol{w}_\perp\|^2} - e^{uw_{\text{sp}}/\|\boldsymbol{w}_\perp\|^2}\Big)du \\
&= -2\varphi(w_{\text{sp}}) \int_A^B e^{-u^2/2\|\boldsymbol{w}_\perp\|^2} \sinh\Big(uw_{\text{sp}}/\|\boldsymbol{w}_\perp\|^2\Big)du
\end{aligned}$$

and so we have

$$|\varepsilon_{\boldsymbol{w}}| \lesssim \varphi(w_{\text{sp}}) \int_A^B \exp\Big(-u^2/2\|\boldsymbol{w}_\perp\|^2\Big)\Big|\sinh\Big(uw_{\text{sp}}/\|\boldsymbol{w}_\perp\|^2\Big)\Big|du + \frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

from here we use that $|\sinh x| \le \cosh x \le \exp(|x|)$ and take a uniform upper bound to get

$$\leq \varphi(w_{\text{sp}})(B-A)\max_{u\in[A,B]}\underbrace{\exp\left(-u^2/2\|\boldsymbol{w}_\perp\|^2\right)}_{\leq 1}\underbrace{\exp\left(\left|\frac{uw_{\text{sp}}}{\|\boldsymbol{w}_\perp\|^2}\right|\right)}_{\text{maxmized at } u=D=|A|}+\frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

$$\leq \varphi(w_{\text{sp}})(B-A)\exp\left(|A|w_{\text{sp}}/\|\boldsymbol{w}_\perp\|^2\right)+\frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

$$=\frac{1}{\sqrt{2\pi}\|\boldsymbol{w}_\perp\|}\exp\left(-w_{\text{sp}}^2/2\|\boldsymbol{w}_\perp\|^2\right)(B-A)\exp\left(|A|w_{\text{sp}}/\|\boldsymbol{w}_\perp\|^2\right)+\frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

from here we use that $B-A\leq|A|$ as A,B,C,D are the four ways we can assign signs to w_1,w_2 with $|A|=D$ being the maximizer, where $B-A\in\{A,B,C,D\}$ by linearity. Furthermore $|A|\leq\sqrt{2}\|\boldsymbol{w}_{1:2}\|$ by Cauchy-Schwarz so we have

$$\lesssim\frac{B-A}{\|\boldsymbol{w}_\perp\|}\exp\left(\frac{2|A|w_{\text{sp}}-w_{\text{sp}}^2}{2\|\boldsymbol{w}_\perp\|_2^2}\right)+\frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

$$\lesssim\frac{\|\boldsymbol{w}_{1:2}\|}{\|\boldsymbol{w}_\perp\|}\exp\left(\frac{\sqrt{2}\|\boldsymbol{w}_{1:2}\|\|\boldsymbol{w}_{\text{sp}}\|-w_{\text{sp}}^2}{\|\boldsymbol{w}_\perp\|_2^2}\right)+\frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|_2^3}$$

as claimed.

■

Lemma 0.4 (L_0 gradient along perp): For any neuron $(a,\boldsymbol{w})$ we have

$$-\boldsymbol{w}_\perp^\top\nabla_{\boldsymbol{w}}L_0=\frac{1}{8}|a|\mathbb{E}\Big[|\boldsymbol{w}^\top\boldsymbol{\xi}|\Big(\mathbb{1}\Big[|\boldsymbol{w}^\top\boldsymbol{\xi}|\geq\left|\sqrt{2}\|\boldsymbol{w}_{\text{sig}}\|+w_{\text{sp}}\right|\Big]+\mathbb{1}\Big[|\boldsymbol{w}^\top\boldsymbol{\xi}|\geq\left|\sqrt{2}\|\boldsymbol{w}_{\text{sig}}\|-w_{\text{sp}}\right|\Big]$$
$$\mathbb{1}\Big[|\boldsymbol{w}^\top\boldsymbol{\xi}|\geq\left|\sqrt{2}\|\boldsymbol{w}_{\text{opp}}\|+w_{\text{sp}}\right|\Big]+\mathbb{1}\Big[|\boldsymbol{w}^\top\boldsymbol{\xi}|\geq\left|\sqrt{2}\|\boldsymbol{w}_{\text{opp}}\|-w_{\text{sp}}\right|\Big]\Big)\Big]$$

Proof of Lemma 0.4:

As before we can use the definition of L_0 to see that

$$-\boldsymbol{w}_\perp^\top\nabla_{\boldsymbol{w}}L_0=-\boldsymbol{w}_\perp^\top\mathbb{E}[-y(\boldsymbol{x})a\sigma'(\boldsymbol{w}^\top\boldsymbol{x})\boldsymbol{x}]=\mathbb{E}[y(\boldsymbol{z})a\sigma'(\boldsymbol{w}^\top\boldsymbol{x})\boldsymbol{w}_\perp^\top\boldsymbol{\xi}]$$

and here we'll appeal to the symmetry of $\boldsymbol{\xi}$ right at the start to get this equals

$$=\frac{1}{2}a\mathbb{E}[y(\boldsymbol{z})\boldsymbol{w}_\perp^\top\boldsymbol{\xi}(\sigma'(\boldsymbol{w}^\top\boldsymbol{z}+\boldsymbol{w}^\top\boldsymbol{s}+\boldsymbol{w}^\top\boldsymbol{\xi})-\sigma'(\boldsymbol{w}^\top\boldsymbol{z}+\boldsymbol{w}^\top\boldsymbol{s}-\boldsymbol{w}^\top\boldsymbol{\xi}))]$$

$$=\frac{1}{2}a\mathbb{E}[y(\boldsymbol{z})\boldsymbol{w}_\perp^\top\boldsymbol{\xi}(\mathbb{1}[\boldsymbol{w}^\top\boldsymbol{z}+\boldsymbol{w}^\top\boldsymbol{s}+\boldsymbol{w}^\top\boldsymbol{\xi}]-\mathbb{1}[\boldsymbol{w}^\top\boldsymbol{z}+\boldsymbol{w}^\top\boldsymbol{s}-\boldsymbol{w}^\top\boldsymbol{\xi}])]$$

as before, notice that $\mathbb{1}[a+b+c]-\mathbb{1}[a+b-c]$ is one when $-c\leq a+b\leq c$ and negative one when $c\leq a+b\leq -c$ and so it equals $\text{sgn}(c)\mathbb{1}[|a+b|\leq|c|]$, and so the above equals, after using $\text{sgn}(x)x=|x|$,

$$=\frac{1}{2}a\mathbb{E}\left[y(z)|\boldsymbol{w}_\perp^\top\boldsymbol{\xi}|\mathbb{1}\left[|\boldsymbol{w}_\perp^\top\boldsymbol{\xi}|\geq|\boldsymbol{w}^\top\boldsymbol{z}+\boldsymbol{w}^\top\boldsymbol{s}|\right]\right]$$

unlike the proof of Lemma 0.1 we're not doing any conditioning on whether $\boldsymbol{w}_{\text{sig}} \| \boldsymbol{z}$ so we just expand here over $(\boldsymbol{z}, \boldsymbol{s})$, grouping $1-\lambda$ terms and λ terms that have the same value because of absolute values to get

$$\begin{aligned}
= \frac{1}{8} a \mathbb{E}\big[|\boldsymbol{w}_\perp^\top \boldsymbol{\xi}| \big(& \\
& \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq |\boldsymbol{w}^\top \boldsymbol{\mu_1} + w_{\text{sp}}|\big] \\
& + \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq |\boldsymbol{w}^\top \boldsymbol{\mu_1} - w_{\text{sp}}|\big] \\
& - \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq |\boldsymbol{w}^\top \boldsymbol{\mu_2} + w_{\text{sp}}|\big] \\
& - \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq |\boldsymbol{w}^\top \boldsymbol{\mu_2} - w_{\text{sp}}|\big] \\
& \big)\big]
\end{aligned}$$

and if $(\boldsymbol{w}, a)$ is a positive neuron then $\boldsymbol{w}^\top \boldsymbol{\mu_1} = \sqrt{2}\|\boldsymbol{w}_{\text{sig}}\|$, and for negative neurons $\boldsymbol{w}_{\text{sig}}, \boldsymbol{w}_{\text{opp}}$ swap roles (so they get opposite signs). Given that negative neurons have $a < 0$ we can compactly write this as

$$\begin{aligned}
= \frac{1}{8} |a| \mathbb{E}\big[|\boldsymbol{w}_\perp^\top \boldsymbol{\xi}| \big(& \\
& \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq \sqrt{2}\|\boldsymbol{w}_{\text{sig}}\| + w_{\text{sp}}\big)\big] \\
& + \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq \sqrt{2}\|\boldsymbol{w}_{\text{sig}}\| - w_{\text{sp}}\big)\big] \\
& - \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq \sqrt{2}\|\boldsymbol{w}_{\text{opp}}\| + w_{\text{sp}}\big)\big] \\
& - \mathbb{1}\big[|\boldsymbol{w}^\top \boldsymbol{\xi}| \geq \sqrt{2}\|\boldsymbol{w}_{\text{opp}}\| - w_{\text{sp}}\big)\big] \\
& \big)\big]
\end{aligned}$$

as desired. **TODO:** why is this expression uglier than the signal direction's expression? Does it say something interesting about how xi and the first two coordinates behave differently? ■

Lemma 0.5 (L_0 gradient for a coordiante of perp): For any neuron $(a, \boldsymbol{w})$ we have

$$\begin{aligned}
-w_i \partial_{w_i} L_0 = \frac{1}{8}|a||w_i| & \\
\Big[\Pr_{\boldsymbol{\xi}}\big[|w_i| \geq \big|\sqrt{2}\|\boldsymbol{w}_{\text{sig}}\| + w_{\text{sp}} + \boldsymbol{w}^\top \boldsymbol{\xi}_{\setminus i}\big|\big] + \Pr_{\boldsymbol{\xi}}\big[|w_i| \geq \big|\sqrt{2}\|\boldsymbol{w}_{\text{sig}}\| - w_{\text{sp}} + \boldsymbol{w}^\top \boldsymbol{\xi}_{\setminus i}\big|\big] & \\
- \Pr_{\boldsymbol{\xi}}\big[|w_i| \geq \big|\sqrt{2}\|\boldsymbol{w}_{\text{opp}}\| + w_{\text{opp}} + \boldsymbol{w}^\top \boldsymbol{\xi}_{\setminus i}\big|\big] - \Pr_{\boldsymbol{\xi}}\big[|w_i| \geq \big|\sqrt{2}\|\boldsymbol{w}_{\text{opp}}\| - w_{\text{opp}} + \boldsymbol{w}^\top \boldsymbol{\xi}_{\setminus i}\big|\big] \Big] &
\end{aligned}$$

Proof of Lemma 0.5: This proof can be done in the same way as Lemma 0.4 just applied coordinate wise, although here I will do a slightly different approach that expands over x_i, using that x_i is Rademacher for $i > 3$, rather than appealing to symmetry which changes the first few steps. Pretty sure that this works but it's late right now so **TODO:** check this

From definition we have that

$$-w_i\partial_{w_i}L_0 = \mathbb{E}_x[ay(\boldsymbol{x})\sigma'(\boldsymbol{w}^\top\boldsymbol{x})w_ix_i]$$
$$= \frac{1}{2}a\mathbb{E}_x[y(z)(\sigma'(\boldsymbol{w}^\top\boldsymbol{x}_{\setminus i}+w_i)w_i - \sigma'(\boldsymbol{w}^\top\boldsymbol{x}_{\setminus i}-w_i)w_i)]$$
$$= \frac{1}{2}aw_i\mathbb{E}_x[y(z)(\sigma'(\boldsymbol{w}^\top\boldsymbol{x}_{\setminus i}+w_i) - \sigma'(\boldsymbol{w}^\top\boldsymbol{x}_{\setminus i}-w_i))]$$

and we can reuse the same difference of indicators analysis to get that this equals

$$= \frac{1}{2}aw_i\mathbb{E}_x\left[y(z)\mathbb{1}\left[|\boldsymbol{w}^\top\boldsymbol{x}_{\setminus i}| \le |w_i|\right]\operatorname{sgn}(w_i)\right]$$
$$= \frac{1}{2}a|w_i|\mathbb{E}_x\left[y(z)\mathbb{1}\left[|w_i| \ge \left|\boldsymbol{w}^\top\boldsymbol{x}_{\setminus i}\right|\right]\right]$$
$$= \frac{1}{2}a|w_i|\mathbb{E}_x\left[y(z)\mathbb{1}\left[|w_i| \ge \left|\boldsymbol{w}^\top\boldsymbol{z} + \boldsymbol{w}^\top\boldsymbol{s} + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right]\right]$$

as in Lemma 0.4 we expand over the distribution $(\boldsymbol{z}, \boldsymbol{\xi})$, using the structure of the absolute value being taken on the right hand side to group pairs of terms with probability $\frac{1}{4}(1-\lambda)$ and $\frac{1}{4}\lambda$ that take matching values under absolute value to get

$$= \frac{1}{2}a|w_i|\mathbb{E}_{\boldsymbol{\xi}}$$
$$\left[\mathbb{1}\left[|w_i| \ge \left|\boldsymbol{w}^\top\boldsymbol{\mu}_1 + \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right] + \mathbb{1}\left[|w_i| \ge \left|\boldsymbol{w}^\top\boldsymbol{\mu}_1 - \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right]\right.$$
$$\left.-\mathbb{1}\left[|w_i| \ge \left|\boldsymbol{w}^\top\boldsymbol{\mu}_2 + \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right] - \mathbb{1}\left[|w_i| \ge \left|\boldsymbol{w}^\top\boldsymbol{\mu}_2 - \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right]\right]$$

and as in Lemma 0.4 we note that when $a > 0$ we take $\boldsymbol{w}_{\text{sig}}$ to align with $\boldsymbol{\mu}_1$, and for $a < 0$ it's with $\boldsymbol{\mu}_2$ and so we can use that the above terms swap $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$'s roles up to signs to write this as

$$= \frac{1}{2}|a||w_i|\mathbb{E}_{\boldsymbol{\xi}}$$
$$\left[\mathbb{1}\left[|w_i| \ge \left|\sqrt{2}\|\boldsymbol{w}_{\text{sig}}\| + \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right] + \mathbb{1}\left[|w_i| \ge \left|\sqrt{2}\|\boldsymbol{w}_{\text{sig}}\| - \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right]\right.$$
$$\left.-\mathbb{1}\left[|w_i| \ge \left|\sqrt{2}\|\boldsymbol{w}_{\text{opp}}\| + \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right] - \mathbb{1}\left[|w_i| \ge \left|\sqrt{2}\|\boldsymbol{w}_{\text{opp}}\| - \boldsymbol{w}_{\text{sp}}\boldsymbol{e}_3 + \boldsymbol{w}^\top\boldsymbol{\xi}_{\setminus i}\right|\right]\right]$$

with the a being in absolute values handling the fact that for $a < 0$ we pick up a negative sign by swapping the roles of $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$. The result then follows by writing the expectation of an indicator as a probability. ■

Section C: L_ρ Analysis

- First shows a bound on the difference in the gradients between L_0 and L_ρ, coordinate by coordinate
- Then we reuse results from section B to characterize the gradients in $\{\boldsymbol{w}_{\text{sig}}, \boldsymbol{w}_{\text{opp}}, \boldsymbol{w}_{\text{sp}}, \boldsymbol{w}_\perp\}$ under L_ρ, but with extra steps to handle the margin term that results from us considering $\ell_\rho^{(1)}$ rather than $\ell_0^{(1)} = -1$. All lemmas rely on a certain assumption on the magnitude of the margin, and condition on a high probability event $\mathcal{E}_{\text{test}}$, generally bounding via a Lipschitzness argument.

Section D: Phase I Induction

Preliminaries

- The first lemma shows that if if we get alignment between the signs of $a, \boldsymbol{w}_{\text{sp}}$ then and $\boldsymbol{w}_{\text{sp}}$ is sufficiently large compared to the other components then the network predicts based on $\boldsymbol{w}_{\text{sp}}$
  - a determines sign, so if it matches $\boldsymbol{w}_{\text{sp}}$ then $\boldsymbol{w}_{\text{sp}}$ determines sign, so we get no contribution when its sign differs from x_3 and otherwise we get a contribution matching x_3

Scalings

1. $\log\log(d)d^{-C} \ll \eta \ll \log^{-3}(d)$
   - upper bound helps control a as being close to $\|\boldsymbol{w}\|$, and lower keeps $t \lesssim d^C$
2. $t \lesssim \log\log(d)\eta^{-1}$
   - length of phase I
3. $\log(d) \ll p$ helps to ensure $\mathcal{E}_{\text{init}}$
   - so norm of $\boldsymbol{w}_\perp$ controlled, and balance of S^+, S^-
4. $\theta \ll \log^{-5C}(d)$
   - limited by the analysis of the per-step change in $\boldsymbol{w}_\perp$
5. $m \gg d\log^{7C}(d)$
   - Limited by Hoeffding's

Inductive Hypothesis

1. $\boldsymbol{w}_{\text{sp}}^{(t)} - \boldsymbol{w}_{\text{sp}}^{(t-1)} = \eta(1 \pm o(\log^{-2}(d))) \cdot \left(\frac{1}{2} - \lambda\right) \cdot \text{sgn}\left(a^{(0)}\right)\left(\left|\boldsymbol{w}_{\text{sp}}^{(t-1)}\right| + \theta\right)$
2. $\left\|\boldsymbol{w}_{\text{sig}}^{(t)} - \boldsymbol{w}_{\text{sig}}^{(t-1)}\right\| \lesssim \eta\left(\left\|\boldsymbol{w}_{\text{sig}}\right\|^{(t-1)} + \theta\log^C(d)d^{-1/2}\right)$
3. $\left\|\boldsymbol{w}_{\text{opp}}^{(t)} - \boldsymbol{w}_{\text{opp}}^{(t-1)}\right\| \lesssim \eta\left(\left\|\boldsymbol{w}_{\text{opp}}\right\|^{(t-1)} + \theta\log^C(d)d^{-1/2}\right)$
4. $\left\|\boldsymbol{w}_\perp^{(t)} - \boldsymbol{w}_\perp^{(t-1)}\right\| \lesssim \eta\theta\log^{-2C}(d)$
5. $\left\|\boldsymbol{w}_\perp^{(t)} - \boldsymbol{w}_\perp^{(t-1)}\right\|_\infty \lesssim \eta\theta\log^{3C}(d)d^{-1/2}$
6. $\left|a^{(t)}\right| \leq \left\|w^{(t)}\right\|$

Phase I Asymptotics

The w part

$$\left\|\boldsymbol{w}_{\text{sp}}^{(t)}\right\| \lesssim \theta\log^C(d) \tag{1.}$$

$$\left\|\boldsymbol{w}_{\text{sig}}\right\|, \left\|\boldsymbol{w}_{\text{opp}}\right\| \lesssim \theta\log^{2C}(d)d^{-1/2} \tag{2.}$$

$$\left\|\boldsymbol{w}_\perp^{(t)}\right\| \asymp \theta, \text{and } \left\|\boldsymbol{w}_\perp^{(t)} - \boldsymbol{w}_\perp^{(0)}\right\| \lesssim \theta\log^{-C}(d) \tag{3.}$$

$$\left\|\boldsymbol{w}_\perp^{(t)}\right\|_\infty \lesssim \theta\log^{4C}(d)d^{-1/2} \tag{4.}$$

$$\left\|w^{(t)}\right\| \asymp \left\|\boldsymbol{w}_{\text{sp}}^{(t)}\right\| + \left\|\boldsymbol{w}_\perp^{(t)}\right\| \lesssim \theta\log^C(d) \tag{5.}$$

The a part

$$\left\|a^{(t)}\right\| \lesssim (1 \pm o(\log^{-2}(d))) \cdot \left\|\boldsymbol{w}^{(t)}\right\| \lesssim (1 \pm o(\log^{-2}(d))) \cdot \left(\left\|\boldsymbol{w}_{\text{sp}}^{(t)}\right\| + \left\|\boldsymbol{w}_\perp^{(t)}\right\|\right) \lesssim \theta\log^C(d) \tag{1.}$$

$$\operatorname{sgn}\big(a^{(t)}\big) = \operatorname{sgn}\big(a^{(0)}\big) \tag{2.}$$

$$\mathbb{E}_{(a,\boldsymbol{w})\sim\rho}\big\|a^{(t)}\boldsymbol{w}^{(t)}\big\| \lesssim \theta^2\log^{2C}(d) \tag{3.}$$

Phase Inductive Steps

w_{sp} inductive step

We want want to show that if the IH holds for $k \leq t$ and we have $\mathcal{E}_{\text{train}}$ which is needed for the lemmas that establish asymptotics, then we have

$$w_{\text{sp}}^{(t+1)} - w_{\text{sp}}^{(t)} = \eta(1 \pm o(\log^{-2}(d)))\left(\frac{1}{2} - \lambda\right)\operatorname{sgn}\big(a^{(0)}\big)\big(|w_{\text{sp}}| + \theta\big)$$

and I will not use the superscript notation to denote the iteration as most of the time it's t or otherwise clear. It's important to note we have $\operatorname{sgn}\big(a^{(0)}\big)$ in the statement though.

We first consider the definition of our SGD update; we have that

$$w_{\text{sp}}^{(t+1)} - w_{\text{sp}}^{(t)} = -\eta\partial_{w_{\text{sp}}}\hat{L}_\rho = -\eta\Big(\partial_{w_{\text{sp}}}L_0 + \partial_{w_{\text{sp}}}\big(\hat{L}_\rho - L_0\big)\Big) = -\frac{\eta}{w_{\text{sp}}}\Big(\boldsymbol{w}_{\text{sp}}^\top\nabla_{\boldsymbol{w}}L_0 + \boldsymbol{w}_{\text{sp}}^\top\nabla_{\boldsymbol{w}}\big(\hat{L}_\rho - L_0\big)\Big)$$

and in Lemma 0.2 we established that

$$-\boldsymbol{w}_{\text{sp}}^\top\nabla_{\boldsymbol{w}}L_0 = aw_{\text{sp}}\left(\frac{1}{2} + \frac{\varepsilon_{\boldsymbol{w}}}{4} - \lambda\right)$$

where we will show that from Lemma 0.3 it follows that $|\varepsilon_{\boldsymbol{w}}| \lesssim \log^{-2}(d)$. From there we use Equation 1 and Equation 3 in the Phase I asymptotics to establish that $\big|a^{(t)}\big| = (1 + o(\log^{-2}(d)))\big(|w_{\text{sp}}| + \theta\big)$ implying

$$-\boldsymbol{w}_{\text{sp}}^\top\nabla_{\boldsymbol{w}}L_0 = (1 \pm o(\log^{-2}(d)))\left(\frac{1}{2} - \lambda\right)\operatorname{sgn}\big(a^{(t)}\big)\operatorname{sgn}(w_{\text{sp}})|w_{\text{sp}}|\big(|w_{\text{sp}}| + \theta\big)$$

we will then show that the $\boldsymbol{w}_{\text{sp}}^\top\nabla_{\boldsymbol{w}}\big(\hat{L}_\rho - L_0\big)$ term is asymptotically dominated such that

$$\begin{aligned}
w_{\text{sp}}^{(t+1)} - w_{\text{sp}}^{(t)} &= -\frac{\eta}{w_{\text{sp}}}\left(-(1 \pm o(\log^{-2}(d)))\left(\frac{1}{2} - \lambda\right)\operatorname{sgn}\big(a^{(t)}\big)\operatorname{sgn}(w_{\text{sp}})|w_{\text{sp}}|\big(|w_{\text{sp}}| + \theta\big)\right) \\
&= \eta(1 \pm o(\log^{-2}(d)))\left(\frac{1}{2} - \lambda\right)\operatorname{sgn}\big(a^{(t)}\big)\big(|w_{\text{sp}}| + \theta\big) \\
&= \eta(1 \pm o(\log^{-2}(d)))\left(\frac{1}{2} - \lambda\right)\operatorname{sgn}\big(a^{(0)}\big)\big(|w_{\text{sp}}| + \theta\big)
\end{aligned}$$

Where we used Equation 2 in the last step. Okay so then it just remains to show the claimed asymptotic bound on $|\varepsilon_{\boldsymbol{w}}|$ and to establish that the $\boldsymbol{w}_{\text{sp}}^\top\nabla_{\boldsymbol{w}}\big(\hat{L}_\rho - L_0\big)$ is asymptotically small. First considering $|\varepsilon_{\boldsymbol{w}}|$ recall that by Lemma 0.3 we have

$$\begin{aligned}
|\varepsilon_{\boldsymbol{w}}| &\lesssim \frac{\|\boldsymbol{w}_{1:2}\|}{\|\boldsymbol{w}_\perp\|}\exp\left(\frac{-\|\boldsymbol{w}_{\text{sp}}\|^2 + 2\sqrt{2}\|\boldsymbol{w}_{1:2}\|\|\boldsymbol{w}_{\text{sp}}\|}{2\|\boldsymbol{w}_\perp\|_2^2}\right) + \frac{\|\boldsymbol{w}_\perp\|_3^3}{\|\boldsymbol{w}_\perp\|^3} \\
&\leq \frac{\|\boldsymbol{w}_{1:2}\|}{\|\boldsymbol{w}_\perp\|}\exp\left(\frac{2\sqrt{2}\|\boldsymbol{w}_{1:2}\|\|\boldsymbol{w}_{\text{sp}}\|}{2\|\boldsymbol{w}_\perp\|_2^2}\right) + \frac{\|\boldsymbol{w}_\perp\|_\infty}{\|\boldsymbol{w}_\perp\|}
\end{aligned}$$

and we plug in our scalings

$$\|\boldsymbol{w}_{1:2}\| \lesssim \big(\theta \log^{2C}(d)\big)/\sqrt{d}$$
$$\|\boldsymbol{w}_{\perp}\|_2 \asymp \theta$$
$$\|\boldsymbol{w}_{\text{sp}}\| \lesssim \theta \log^{C}(d)$$
$$\|\boldsymbol{w}_{\perp}\|_\infty \lesssim \theta \log^{4C}(d)/\sqrt{d}$$

to get that the above is

$$\lesssim \frac{\theta \log^{2C}(d)/\sqrt{d}}{\theta} \exp\left(\frac{C\theta \log^{2C}(d) \cdot \theta \log^{C}(d)}{\theta^2\sqrt{d}}\right) + \frac{\theta \log^{4C}(d)}{\theta\sqrt{d}}$$
$$= \log^{2C}/\sqrt{d} \cdot \exp\big(C \log^{3C}(d)/\sqrt{d}\big) + \log^{4C}(d)/\sqrt{d}$$

and use that $1 + u \lesssim \exp(u)$ (yes this isn't meant to be $1 + x \leq \exp(x)$) to get

$$\lesssim \frac{1}{\sqrt{d}}\big(\log^{2C}\big(1 + \log^{3C}(d)\big) + \log^{4C}(d)\big) \lesssim \log^{5C}(d)/\sqrt{d} \lesssim \log^{2C}(d)$$

as desired. Now considering $-\boldsymbol{w}_{\text{sp}}^\top \nabla_{\boldsymbol{w}}\big(\hat{L}_\rho - L_0\big)$ observe that **TODO:** type up this lemma at some point and ref in the $i = 3$ case states

$$\big|\partial_{\boldsymbol{w}_{\text{sp}}}\big(\hat{L}_\rho - L_0\big)\big| \lesssim \theta^3 \log^{4C}(d) + \theta \log^{-2}(d)/\sqrt{d}$$

and so

$$\begin{aligned}\big|\boldsymbol{w}_{\text{sp}}^\top \nabla_{\boldsymbol{w}}\big(\hat{L}_\rho - L_0\big)\big| &\leq \|\boldsymbol{w}_{\text{sp}}\| \cdot \big\|\boldsymbol{e}_3^\top \nabla_{\boldsymbol{w}}\big(\hat{L}_\rho - L_0\big)\big\| \\ &= |w_{\text{sp}}| \cdot \big\|\partial_{\boldsymbol{w}_{\text{sp}}}\big(\hat{L}_\rho - L_0\big)\big\| \\ &\lesssim |w_{\text{sp}}|\big(\theta^3 \log^{4C}(d) + \theta \log^{-2}(d)/\sqrt{d}\big)\end{aligned}$$

our scalings assume that $\theta \ll \log^{-5C}(d)$ so this is

$$\lesssim |w_{\text{sp}}|\big(\theta \log^{-6C}(d) + \theta \log^{-2}(d)/\sqrt{d}\big) \lesssim |w_{\text{sp}}|\theta \lesssim \boldsymbol{w}_{\text{sp}}^\top \nabla_{\boldsymbol{w}} L_0$$

as claimed.